The empirical recurrence for OEIS A281932, recovered from its tail

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Abstract

OEIS A281932 records "Empirical recurrence of order 69" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 754$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 75$. Its last coefficient is $c_{69} = 110592 \neq 0$, so no further tail satisfies anything shorter; any order-69 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A281932 is "Number of nX4 0..1 arrays with no element equal to more than four of its king-move neighbors, with the exception of exactly two elements, and with new values introduced in order 0 sequentially upwards.". It has offset 1 and begins

$$0, 6, 312, 6304, 105766, 1488168, 19173138, 233189094, \ \ldots$$

The entry states:

> Empirical recurrence of order 69 (see link above)

As of the "Last modified" line on the live entry (Feb 02 2017 08:20:52, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 754$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 754$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 69: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 69.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1508$ exact terms from $n = 6$ on, it returns order 69 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{69} c_i \, a(n-i),$$

c_1	27	c_{19}	−599557971	c_{37}	−74180801259	c_{55}	915194853
c_2	−243	c_{20}	−8229459912	c_{38}	56910345666	c_{56}	505491594
c_3	825	c_{21}	−13360672604	c_{39}	4741128213	c_{57}	−258243165
c_4	−2484	c_{22}	−3523446552	c_{40}	29246299416	c_{58}	−240705018
c_5	20298	c_{23}	21460852929	c_{41}	−60217172643	c_{59}	−1542660
c_6	−45714	c_{24}	33601038952	c_{42}	5009251147	c_{60}	16711194
c_7	7419	c_{25}	11884804332	c_{43}	−23060340849	c_{61}	40798428
c_8	−666945	c_{26}	−38252596101	c_{44}	35434784292	c_{62}	16312200
c_9	706106	c_{27}	−43738664996	c_{45}	−3250021779	c_{63}	−11229724
c_{10}	3027171	c_{28}	2352690459	c_{46}	8568272394	c_{64}	−12099576
c_{11}	14368980	c_{29}	55731110523	c_{47}	−17342886180	c_{65}	3977952
c_{12}	4552515	c_{30}	20014793916	c_{48}	6695941506	c_{66}	3131000
c_{13}	−69603162	c_{31}	−63358962942	c_{49}	1358551257	c_{67}	−1105344
c_{14}	−227427630	c_{32}	−58423068867	c_{50}	5406026985	c_{68}	−235008
c_{15}	−269385362	c_{33}	19950021319	c_{51}	−3154090418	c_{69}	110592
c_{16}	442151511	c_{34}	110423926431	c_{52}	−1173607200		
c_{17}	1866695670	c_{35}	8681375112	c_{53}	−2019605013		
c_{18}	2803696185	c_{36}	−29956301907	c_{54}	436362860		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 75$, and it is the recurrence OEIS A281932 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{69} - c_1 x^{69-1} - \cdots - c_{69}$. Its constant term is $-c_{69} = -110592$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with

$\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 69 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 69, so $\deg q = 69$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 69, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 69, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 110592.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-69 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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